GOOD TRUNCATIONS FOR COMPUTING INTEGRAL BASES

Abstract. We provide good truncation bound that speed up the computation of integral basis.

1. Introduction

Integral basis are very useful in real life.

Given a polynomial $f \in K[x, y]$ monic in Y, we wish to compute the local contribution to the integral basis at the origin or the local integral basis at the origin.

2. The determinacy

In this section we show:

(1) An upper bound for the determinacy can be found fast reducing the polynomial modulo any prime number p.
(2) If we truncate the polynomial f by standard degree at the determinacy, the characteristic exponents of f will not change.
(3) The integrality exponent of f can be computed from the characteristic exponents, so we can compute very fast the integrality exponent.

Remark: the coefficients of the Puiseux expansions might change, but in all examples we tried, the coefficients don't change after truncation.

3. Known results about precision of factorization and computation of Puiseux expansions

We recall first two results that will be useful for our method.

The first result is about factorization over $K[[X]][Y]$.

Proposition 3.1. *(See [2, Proposition 23].) Let $F \in K[[X]][Y]$, separable of degree d. Suppose given a modular factorisation*

$$F \equiv F_0 F_1 \dots F_s \mod X^{N_0},\ N_0 > 2\kappa \tag{1}$$

where $F_0 \in K[[X]][Y]$ is a unit and for all $i \geq 1$ either F_i or its reciprocal polynomial $\tilde{F}_i$ is monic, and

$$\kappa = \kappa(F_0, ..., F_s) := \max_{I,J} \kappa(F_I, F_J),$$

the maximum of the lifting orders being taken over all disjoint subsets $I, J \subset 0, \dots, s$, with $F_I = \prod_{i\in I} F_i$. Then there exists uniquely determined analytic factors $F_0^*, F_1^*, \dots, F_k^*$ such that $F = F_0^* F_1^* \cdots F_k^*$, where

$$F_i^* \equiv F_i \mod X^{N_0' - \kappa}.$$

Moreover, starting from (1), we can compute the $F_i^\star$ up to any precision $N \geq N_0 - \kappa$ in $\mathcal{O}(dN)$ operations over K.

The authors provide the bound $\kappa \leq \delta/2$, where $\delta = v_X\left(\mathrm{Res}_Y(F, F_Y)\right) = 2\mathrm{IntExp}_X$, so $\kappa \leq \mathrm{IntExp}_X$.

The second result provides bound for the precision required for computing Puiseux expansions. We use this algorithm taking $\mathbb{K}_P = \mathbb{Q}$.

We denote by $(R_i)_{1\leq i\leq \rho}$ the rational Puiseux expansions of H. To any R_i, we associate the following notations:

- e_i is the ramification index (equal to the largest denominator or the number of expansions in the branch),
- f_i is the residual degree,
- the *regularity index* r_i of a Puiseux series R_i of F with ramification index e_i is the least integer $N \geq \min(0, ev_X(R_i))$ such that, if
$$\lceil R_i \rceil^{N/e_i} = \lceil R' \rceil^{N/e_i}$$
for some Puiseux series R' of F, then $R_i = R'$. We call $\lceil R_i \rceil^{r_i/e_i}$ the *singular part* of R_i in F.
- we define $v_i \in \mathbb{Q}$ as $v_X(F_Y(S))$, for any Puiseux series S associated to R_i.

  If H has only only one branch, then this number is the integrality exponent. Proof: if $F_i = (Y-\gamma_1)\ldots(Y-\gamma_r)$ then $(F_i)_Y(\gamma_1) = (1)(\gamma_1-\gamma_2)\cdots(\gamma_1-\gamma_{e_i})+0+0+\cdots+0$. The order of the product is equal to IntExp_X (it is the same formula).

  If H has many branches, then v_i can be bigger than the integrality exponent.

Algorithm 1 Half-RNP(H, P, n, π)

1: **Input:** $P \in \mathbb{K}[Z]$ irreducible; $H \in \mathbb{K}_P[X, Y]$ separable and monic in Y, with $d := \deg_Y(H) > 0$; $n \in \mathbb{N}$ (truncation order); π the current truncated parametrisation (initial call: $P = Z$, $\pi = (X, Y)$)

2: **Output:** All RPEs R_i of H such that $n - v_i \geq r_i$, with precision $(n - v_i)/e_i \geq r_i/e_i$

There seems to be an error in the bounds here, which don't agree with the following theorem in that same paper:

Lemma 3.2. *([2, Lemma 3.19]) Let $n_0 \in \mathbb{N}$. To compute the RPE R_i with certified precision $n_0 > \frac{r_i}{e_i}$, it is necessary and sufficient to run* Half-RNP *with truncation bound $n = n_0 + v_i$. In particular, to ensure the computation of the singular part of R_i, it is necessary and sufficient to use a truncation bound $n > N_i$.*

We use the bounds in the Lemma, which have been confirmed by the authors and by computational experiments.

To give general bounds for the precision n required we define a new number:
$$\alpha = \sum r_i,$$
the sum taken over all Puiseux branches. This number is bigger than the integrality exponent but not much bigger. For example, if we have only one

branch with seven Puiseux expansions and one characteristic exponent $1/7$, then $\alpha = 1$ and the integrality exponent is $6/7$.

We have:

- $r_i \leq \alpha$ (clear by definition)
- $v_i \leq \mathrm{IntExp}_X \leq \alpha$ when there is only one branch, but not in general.

Based on the last observation, we define $\tilde{v}_i = v_X((F_i)_Y(S))$, for any Puiseux series S associated to R_i. This number will be used when we apply Algorithm 1 to only one branch of F.

4. Local contribution to the integral basis at the origin

In this section we focus on the computation of the local contribution at the origin (that is, we want to keep the unit too).

The steps are now as follows:

(1) Proposition 3.1 says that if we truncate F at precision n in X and we factorize the truncated polynomial $\bar{F}$ in $K[[X]][Y]$, we will get the factors of F with precision $n - \kappa$.
(2) We are interested in computing the factors with precision at least equal to the integrality exponent in X of F, we call this number IntExp_X. That is, we need $n - \kappa \geq \mathrm{IntExp}_X$.
(3) In that paper the authors provide the bound $\kappa \leq \delta/2$, where $\delta = v_X\left(\mathrm{Res}_Y(F, F_Y)\right) = 2\mathrm{IntExp}_X$, so $\kappa \leq \mathrm{IntExp}_X$ and it is enough for us to consider $n \geq 2\mathrm{IntExp}_X$.
(4) For the computation of the integral basis we also need the singular part of the Puiseux expansions. If we want to compute the Puiseux expansions from the factors $F_1, \dots, F_s$, we would need to compute the factors with precision higher than the integrality exponent.

    However if we compute the Puiseux expansions from $\bar{F}$, the same proposition 3.1 says that we will obtain the Puiseux expansions with precision at least IntExp_X, and hence the singular part will be correctly computed.
(5) I am not sure about this last step. In any case, we can compute the singular part of the Puiseux expansions by truncating at $\alpha + 2 \times \mathrm{IntExp}_X$ (a little larger than $3 \times \mathrm{IntExp}_X$). The argument is as follows:
    (a) We apply Algorithm 1 Half-RNP to one factor F_i.
    (b) In this case $v_i = v_X((F_i)_Y)$ and this is smaller than IntExp_X (the integrality exponent of one branch is smaller than that of the full polynomial).
    (c) r_i may not be smaller than the integrality exponent but we have $r_i \leq \alpha$.
    (d) So by Algorithm 1, we need $n \geq v_i + r_i$, and $v_i + r_i \leq \alpha + \mathrm{IntExp}_X$. That is, we need the factor F_i developed to degree $\alpha + \mathrm{IntExp}_X$ and by Proposition 3.1 we can get this by computing a factorization of F modulo $\alpha + 2\mathrm{IntExp}_X$.
(6) Putting everything together, we can truncate the powers of X of F at order $\alpha + 2\mathrm{IntExp}_X$ and we will be able to compute the correct factors up to the integrality exponent and the singular part of the

Puiseux expansions and hence we will compute the correct integral basis.

5. Local integral basis at the origin

When we are interested in the local integral basis at the origin we can also truncate powers of Y.

One possible approach is the following:

(1) Applying again [2, Proposition 23] in terms of Y with high enough precision, we obtain that the factors at the origin are uniquely determined and they are the same factors as the factors obtained when we develop the factorization in terms of X, so we are able to recover the correct information.
(2) The factor g_0 however corresponds to the branches of f at $Y = 0$ and the factor f_0 corresponds to the branches of f at $X = 0$, so we cannot recover the Puiseux expansions of f_0 from the Puiseux expansions of g_0.
(3) Hence when we apply a truncation in Y we can expect to compute correctly the local integral basis but not the local contribution.

We now formalize these ideas and calculate the required precision. For this, we need to be able to go from Puiseux expansions in X to Puiseux expansions in Y (N_0 and κ are the numbers to be determined, and they are possibly different for X-truncations and Y-truncations).

(1) We start with a polynomial $F \in K[X, Y]$ with analytic factorization in $K[[X]][Y]$:
$$F = F_0^* F_1^* \dots F_s^*,$$
where $F_k^* \in K[[X]][Y]$ and F_0^* is the unit corresponding to the branches outside the origin). Let d_k be the degree in Y of F_k^* and let e_k be the largest denominator of the Puiseux expansions of F_k^*.
(2) If we have a factorization
$$F \equiv F_0 F_1 \dots F_s \mod X^{N_0}, \tag{2}$$
then by Proposition 3.1, $F_k \equiv F_k^* \mod X^{N_0 - \kappa}$.
(3) Now we truncate in Y. Let $\overline{F}^Y$ be the truncation of F modulo Y^{N_0}. We can factorize this polynomial in $K[[X]][Y]$ as
$$\overline{F}^Y \equiv H_0 H_1 \dots H_s \mod X^{N_0},$$
and we want to know if $H_k \equiv F_k^* \mod X^{N_0-\kappa}$ for $k > 0$ (we cannot expect $H_0 \equiv F_0^* \mod X^{N_0-\kappa}$).
(4) We can also factorize $\overline{F}^Y$ in $K[[Y]][X]$:
$$\overline{F}^Y \equiv G_0 G_1 \dots G_s \mod Y^{N_0}.$$
(5) Applying Proposition 3.1 to F in $K[[Y]][X]$ we obtain that
$$G_k \equiv G_k^* \mod Y^{N_0-\kappa}$$
where G_k^* are the analytic factors of F in $K[[Y]][X]$.

By the same argument G_k agrees with the analytic factors of $\overline{F}^Y$ mod $Y^{N_0-\kappa}$. Hence: the analytic factors in $K[[Y]][X]$ of F and $\overline{F}^Y$ are the same modulo $Y^{N_0-\kappa}$.

The question is what precision do we get when we go from these factors to factors in $K[[X]][Y]$.

(6) Now the key argument is: for fixed $1 \leq k \leq s$, there is a reparametrization (or coordinates change) $(\alpha(X,Y), \beta(X,Y))$ that will transform G_k^*, into the corresponding analytic factor F_k^* of $F \in K[[X]][Y]$, that is

$$G_k^*(\alpha(X,Y), \beta(X,Y)) = F_k^*(X,Y).$$

Modulo $X^{N_0-\kappa}$, we obtain from (2):

$$G_k^*(\alpha(X,Y), \beta(X,Y)) \equiv F_i(X,Y) \mod X^{N_0-\kappa}.$$

We have to show that if we apply this same reparametrization to G_k we will obtain

$$G_k(\alpha(X,Y), \beta(X,Y)) \equiv H_k \equiv F_k \mod X^{N_0-\kappa}.$$

(7) By [1, Theorem 5.17], we can write

$$G_k(x,y) = \prod_{i=1}^{n}(x - \eta(\zeta^i y^{1/n})) = \prod_{i=1}^{n}(x - \eta(\zeta^i u)),$$

where ζ is a primitive n–th root of unity and $y = u^n$. The corresponding parametrization of the singularity is

$$(\eta(u), u^n)$$

and we want to compute the parametrization $(t^m, \gamma(t))$.

That is, we want to invert th function $x = \eta(u)$ to obtain $y = \gamma(t)$.

So let

$$\eta(u) = (a_0 u^{m_0} + a_1 u^{m_0+1} + a_2 u^{m_0+2} + \dots,$$

$a_0 \neq 0$ (here $n = e_k$ and $m_0 = d_k$).

To invert this transformation we note

$$\begin{aligned}\eta(u) &= u^{m_0}(a_0 + a_1 u + a_1 u^2 + \dots) \\ &= \left(u \sqrt[m_0]{a_0 + a_1 u + a_2 u^2 + \dots}\right)^{m_0}\end{aligned}$$

Let

$$b_0 + b_1 u + b_2 u^2 + \dots = \sqrt[m_0]{a_0 + a_1 u + a_2 u^2 + \dots}$$

be an m_0-th root of $a_0 + a_1 u + a_2 u^2 + \dots$. In computations, if $a_0 + a_1 u + a_1 u^2 + \dots$ is developed up to precision c, then also $b_0 + b_1 U + b_2 U^2 + \dots$ can be computed with precision c.

Set $q(u) = u(b_0 + b_1 u + b_2 u^2 + \dots) = b_0 u + b_1 u^2 + b_2 u^3 + \dots$ (hence $\eta(u) = q(u)^{m_0}$). Again, $q(u)$ can be computed with precision c.

Finally let $p(t) = c_1 t + c_2 t^2 + \dots$ be the composite inverse of $q(u)$ (which can also be computed with precision c), that is $q(p(t)) = t$ and hence $u = p(t)$.

We obtain the new parametrization

$$(t^{m_0}, p(t)^n).$$

As we have seen, all these transformations can be done without precision loss, hence if the x-Puiseux expansions of $G_i^*(x,y)$ are known with precision c/n, then the y-Puiseux expansions of $F_i^*(x,y)$ are known with precision c/m.

Remark 5.1. The argument for truncations in Y is the following:

(1) Let $\bar{F}$ be the truncation of F at some degree $K \geq \alpha_Y + 2\mathrm{IntExp}_Y$ in Y. We want to know a good K.

(2) By Proposition 3.1 we can compute the factors in $K[[X]][Y]$ correct up to degree $n = K - \mathrm{IntExp}_Y \geq \alpha + \mathrm{IntExp}_Y$. Consider the i-th branch. $n - \tilde{v}_i \geq n - \mathrm{IntExp}_Y = K - 2\mathrm{IntExp}_Y \geq \alpha \geq r_i$. Then by Algorithm 1, we can compute the Y-Puiseux expansions of this branch correct up to degree $n - \tilde{v}_i$. Let $D = n - \tilde{v}_i$.

(3) Assume that the Puiseux expansions in X of the i-th branch have denominator ϵ_i (the Puiseux expansions in Y have denominator e_i). Then the polynomial $\eta(U)$ corresponding to that branch is correct up to degree De_i: the Puiseux expansion $\eta(Y^{1/e_i})$ is correct to degree D and we take $U = Y^{1/e_i}$.

(4) Applying the inverse formula we get a polynomial $\gamma(T)$ which is correct up to the same degree De_i in T. Now we replace $T = X^{1/\epsilon_i}$ to get the Puiseux expansions in X, $\gamma(X^{1/\epsilon_i})$. We get correct Puiseux expansions up to degree De_i/ϵ_i.

(5) We need this number De_i/ϵ_i to be bigger than IntExp_X. So we get $D \geq \frac{\epsilon_i}{e_i}\mathrm{IntExp}_X$ and then $K - \mathrm{IntExp}_Y - \tilde{v}_i \geq \frac{\epsilon_i}{e_i}\mathrm{IntExp}_Y$. So we get $K \geq \frac{\epsilon_i}{e_i}\mathrm{IntExp}_X + \mathrm{IntExp}_Y + \tilde{v}_i$.

(6) Consider $\frac{\epsilon}{e}$ the largest $\frac{\epsilon_i}{e_i}$ over all branches. So we need
$$K > \frac{\epsilon}{e}\mathrm{IntExp}_X + 2\mathrm{IntExp}_Y,$$
which is the bound we were looking for.

6. Timings

We get very good timings.

References

[1] Theo de Jong and Gerhard Pfister. *Local analytic geometry. Basic theory and applications.* Braunschweig: Vieweg, 2000.

[2] Adrien Poteaux and Martin Weimann. Computing puiseux series: a fast divide and conquer algorithm. *Annales Henri Lebesgue*, 4:1061–1102, 2021.